

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Integrates using integration by parts	AO3.1a	M1	$y = \int (x-1)e^x dx$
Applies integration by parts formula correctly to either of $(x-1)e^x$ or xe^x	AO1.1a	M1	$u = x-1 \quad \frac{dv}{dx} = 1$ $\frac{dv}{dx} = e^x \quad v = e^x$
Obtains fully correct integral, condone missing constant.	AO1.1b	A1	$y = (x-1)e^x - \int e^x dx$ $y = (x-1)e^x - e^x + c$
Explains clearly why the minimum y value is e with reference to the range of the function OE	AO2.4	E1	Range $\geq e \Rightarrow$ at min $y = e$ Min point when $\frac{dy}{dx} = 0 \therefore x = 1$
Uses $\frac{dy}{dx} = 0$ to find x coordinate of minimum	AO1.1a	M1	So curve passes through $(1, e)$ $e = (1-1)e^1 - e^1 + c$
Deduces that the curve passes through the point $(1, e)$	AO2.2a	A1	$c = 2e$
Uses their minimum point to find their c	AO1.1a	M1	$\therefore f(x) = (x-2)e^x + 2e$
States the correct equation in any correct form Condone y instead of $f(x)$ CAO	AO1.1b	A1	
Total 8 marks			

Q2.

Marking Instructions	AO	Marks	Typical Solution
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(a)	Recalls a correct trig identity, which could lead to a correct answer	AO1.2	B1	(LHS $\equiv$) $\frac{\sin 2x}{1 + \tan^2 x}$
	Demonstrates a strategy for proving the identity, eg by converting all the terms on the LHS to cos and sin.	AO3.1a	M1	$\frac{2 \sin x \cos x}{1 + \tan^2 x}$ $\frac{2 \sin x \cos x}{\sec^2 x}$
	Concludes a rigorous mathematical argument to prove given identity AG	AO2.1	R1	$\equiv 2 \sin x \cos x \cos^2 x$ $\equiv 2 \sin x \cos^3 x$ ($\equiv$ RHS)
(b)	Uses identity to write integrand in the form $a \sin 2\theta \cos^3 2\theta$	AO1.1a	M1	$\int \frac{4 \sin 4\theta}{1 + \tan^2 2\theta} d\theta = \int 8 \sin 2\theta \cos^3 2\theta d\theta$ Let $u = \cos 2\theta$
	Correctly writes integrand as $8 \sin 2\theta \cos^3 2\theta$	AO1.1b	A1	then $\frac{du}{d\theta} = -2 \sin 2\theta \Rightarrow \sin 2\theta = -\frac{1}{2} \frac{du}{d\theta}$ $I = -4 \int u^3 \frac{du}{d\theta} d\theta$
	Selects an appropriate method for integrating, e.g. substitution $u = \cos 2\theta$, or by inspection PI by sight of $\cos^4 2\theta$	AO3.1a	M1	$= -4 \int u^3 du$ $= -u^4 + c$ $= -\cos^4 2\theta + c$
	Obtains $k \int u^3 du$ correctly PI by solution in form $k \cos^4 2\theta$, if by inspection	AO1.1a	M1	
	Obtains $-u^4$ or $-\cos^4 2\theta$ OE Only FT value of a	AO1.1b	A1F	
	Completes rigorous argument to obtain $-\cos^4 2\theta + c$ OE	AO2.1	R1	
				Total 9 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States the correct derivative	AO1.1b	B1	$2^x \ln 2$
(b)	Selects an appropriate	AO3.1a	M1	Let $u = 2^x$

method for integrating, which could lead to a correct exact solution (this could be indicated by an attempt at a substitution or attempting to write the integrand in the form $f(x)f(x)^n$)			$\frac{du}{dx} = 2^x \ln 2$ $\text{Then } \frac{1}{\ln 2} \frac{du}{dx} = 2^x$ $\text{And } \frac{1}{\ln 2} \frac{du}{dx} = 2^x$ $I = \int (3+u)^{\frac{1}{2}} \frac{1}{\ln 2} \frac{du}{dx} dx$ $= \frac{1}{\ln 2} \int (3+u)^{\frac{1}{2}} du$ $= \frac{3}{3\ln 2} (3+u)^{\frac{3}{2}} + c$
Correctly writes integrand in a form which can be integrated (condone missing or incorrect limits)	AO1.1b	A1	Sub limits:
Integrates 'their' expression (allow one error)	AO1.1a	M1	$\left[\frac{2}{3\ln 2} (3+u)^{\frac{3}{2}} \right]_1^2$
Substitutes correct limits corresponding to 'their' method	AO1.1a	M1	$\frac{3}{2} \times \frac{1}{\ln 2} (5\sqrt{5} - 8)$
Obtains correct value in an exact form	AO1.1b	A1	ALT (direct inspection) $\int 2^x \sqrt{3+2^x} dx$ $= \frac{1}{\ln 2} \int 2^x \ln 2 \sqrt{3+2^x} dx$ $= \frac{1}{\ln 2} \int 2^x \ln 2 (3+2^x)^{\frac{1}{2}} dx$ $= \frac{1}{\ln 2} \times \frac{2}{3} (3+2^x)^{\frac{3}{2}}$ $\left[\frac{1}{\ln 2} \times \frac{2}{3} (3+2^x)^{\frac{3}{2}} \right]_0^1$ $\frac{3}{2} \times \frac{1}{\ln 2} (5\sqrt{5} - 8)$
Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips Substitution should be clearly stated in exact form and change of variable or solution by direct inspection should be achieved correctly with correct use of symbols and connecting language	AO2.1	R1	
			Total 7 marks

Q4.

Marking Instructions	AO	Marks	Typical Solution
Selects a method of integration, which could lead to a correct solution. Evidence of integration by parts	AO3.1a	M1	$u = \ln 2x; \quad \frac{dv}{dx} = x^3$ $\frac{du}{dx} = \frac{1}{x}; \quad v = \frac{x^4}{4}$

OR an attempt at integration by inspection.			$\left[\frac{x^4}{4} \ln(2x) \right]_1^2 - \int_1^2 \frac{x^3}{4} dx$
Applies integration by parts formula correctly OR correctly differentiates an expression of the form $Ax^4 \ln 2x$	AO1.1b	A1	$\left[\frac{x^4}{4} \ln(2x) - \frac{x^4}{16} \right]_1^2$ $= \left(\frac{2^4}{4} \ln(4) - \frac{2^4}{16} \right) - \left(\frac{1}{4} \ln(2) - \frac{1}{16} \right)$
Obtains correct integral, condone missing limits.	AO1.1b	A1	$\frac{31}{4} \ln 2 - \frac{15}{16}$
Substitutes correct limits into 'their' integral	AO1.1a	M1	$p = \frac{31}{4} \quad q = -\frac{15}{16}$
Obtains correct p and q FT use of incorrect integral provided both M1 marks have been awarded	AO1.1b	A1F	so $p = \frac{31}{4} \quad q = -\frac{15}{16}$ ALT $\frac{d}{dx}(x^4 \ln 2x) = 4x^3 \ln 2x + x^4 \cdot \frac{1}{x}$ $\therefore \int_1^2 x^3 \ln 2x dx = \left[\frac{1}{4} (x^4 \ln 2x - \frac{x^4}{4}) \right]_1^2$ $= \left(\frac{2^4}{4} \ln(4) - \frac{2^4}{16} \right) - \left(\frac{1}{4} \ln(2) - \frac{1}{16} \right)$ $\frac{31}{4} \ln 2 - \frac{15}{16}$ $p = \frac{31}{4} \quad q = -\frac{15}{16}$
			Total 5 marks

Q5.

(a) $y = 4x \cos 2x$

$$\left(\frac{dy}{dx} = \right) 4 \cos 2x - 4x(2) \sin 2x$$

anything reducible to $A \cos 2x + Bx \sin 2x$ where A and B are non-zero integers

M1

gradient of the tangent

OE, all correct

A1

$$A \cos \frac{2\pi}{4} + B \times \frac{\pi}{4} \sin \frac{2\pi}{4}$$

substituting $\frac{\pi}{4}$ into candidate's derived function

m1

$$= -2\pi$$

must have $= -2\pi$ using correct $\frac{dy}{dx}$

A1

an equation of the tangent is $y = -2\pi \left(x - \frac{\pi}{4} \right)$

OE, dependent on previous A1

A1

5

(b)

$$\left(\int_0^{\frac{\pi}{4}} 4x \cos 2x \, dx \right)$$

$$u = Ax$$

$$\frac{dv}{dx} = \cos 2x$$

$$\frac{du}{dx} = A$$

$$v = B \sin 2x$$

all 4 terms in this form seen or used

M1

$$A = 4 \text{ and } B = \frac{1}{2} \text{ or } A = 1 \text{ and } B = 2, \text{ etc}$$

A1

$$= \left[4x \frac{1}{2} \sin 2x \right]_{(0)}^{\left(\frac{\pi}{4}\right)} - \int_{(0)}^{\left(\frac{\pi}{4}\right)} 4 \times \frac{1}{2} \sin 2x \, (dx)$$

correct substitution of candidate's terms into integration by parts formula condone missing limits

m1

$$= \left[4x \frac{1}{2} \sin 2x \right]_{(0)}^{\left(\frac{\pi}{4}\right)} - \left[-\cos 2x \right]_{(0)}^{\left(\frac{\pi}{4}\right)}$$

*candidate's second integration completed correctly
FT on one error including coefficient condone missing limits*

A1F

$$= \frac{\pi}{2} - 1$$

OE, exact value

A1

5

[10]

Q6.

(a) $\int e^{1-2x} dx = ke^{1-2x}$ or $e(ke^{-2x})$

where k is a rational number

M1

$$\int_0^{\ln 2} e^{1-2x} dx = -\frac{1}{2} e^{1-2x} \Big|_0^{\ln 2} \text{ or } e \left[-\frac{1}{2} e^{-2x} \right]_0^{\ln 2}$$

*correct integration
condone missing limits*

A1

$$= -\frac{1}{2} e^{1-2\ln 2} - -\frac{1}{2} e^{1-2(0)}$$

correct (no decimals)

A1

$$= -\frac{1}{2} \left(\frac{1}{4} e \right) + \frac{1}{2} e$$

eliminating ln

$$= \frac{3}{8} e$$

AG, be convinced

A1

4

(b) $u = \tan x$

$$\frac{du}{dx} = \sec^2 x$$

PI below, condone $du = \sec^2 x dx$

M1

Replacing dx by $\frac{1}{\sec^2 x} (du)$ in integral

$$\text{or } \frac{1}{1+u^2} (du)$$

A1

$$\sec^2 x = 1 + u^2$$

PI below

B1

$$\left. \begin{array}{l} x=0 \Rightarrow u=0 \\ x=\frac{\pi}{4} \Rightarrow u=1 \end{array} \right\}$$

this could be gained by changing u to $\tan x$ after the integration and using $x=0$ and $x=\frac{\pi}{4}$

B1

$$\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt{\tan x} dx$$

$$= \int (1+u^2) \sqrt{u} (du) \text{ or } \int (1+u^2)^2 \sqrt{u} \frac{(du)}{1+u^2}$$

*all in terms of u including replacing dx
all correct, condone omission of du*

M1

$$= \int \left(u^{\frac{5}{2}} + u^{\frac{1}{2}} \right) (du)$$

must be in this form

A1

$$= \frac{2}{7} u^{\frac{7}{2}} + \frac{2}{3} u^{\frac{3}{2}}$$

accept correct unsimplified form

A1

$$= \frac{20}{21}$$

CAO

A1

8

[12]

Q7.

(a) $\int x(x^2 + 3)^{\frac{1}{2}} dx = p(x^2 + 3)^{\frac{3}{2}}$

By inspection or substitution

M1

$$= \frac{1}{3}(x^2 + 3)^{\frac{3}{2}} (+C)$$

A1

2

(b) $\int e^{2y} dy = \int x\sqrt{x^2 + 3} dx$

Correct separation and notation

Condone missing integral signs

B1

$$= \frac{1}{2}e^{2y}$$

B1

$$= \frac{1}{3}(x^2 + 3)^{\frac{3}{2}} + C$$

Equate to result from (a) with constant.

M1

$$\frac{1}{2} = \frac{1}{3} \times 4^{\frac{3}{2}} + C$$

Use (1,0) to find constant.

m1

$$C = -\frac{13}{6}$$

CAO

A1

$$2y = \ln \left(\frac{2}{3}(x^2 + 3)^{\frac{3}{2}} - \frac{13}{3} \right)$$

Solve for y , taking logs correctly.

m1

$$y = \frac{1}{2} \ln \left(\frac{2}{3}(x^2 + 3)^{\frac{3}{2}} - \frac{13}{3} \right)$$

CSO

A1

7

[9]

Q8.

(a) (i)

$$\left. \begin{array}{l}
 u = \ln x \quad \frac{dv}{dx} = 1 \\
 \frac{du}{dx} = \frac{1}{x} \quad v = x
 \end{array} \right\}$$

$\frac{d \ln x}{dx}$ & $\int dx$ attempted

All correct

M1

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A1

$$\left(\int \ln x \, dx = \right) x \ln x - \int x \times \frac{1}{x} (dx)$$

Correct substitution of their terms into parts

m1

$$= x \ln x - x + C$$

All correct (**constant needed**)

A1

4

(ii)

$$\left. \begin{array}{l} u = (\ln x)^2 \\ \frac{dv}{dx} = \end{array} \right\} \\
 \left. \begin{array}{l} \frac{du}{dx} = (2 \ln x) \frac{1}{x} \\ v = \end{array} \right\}$$

$$\frac{d(\ln x)^2}{dx} \& \int dx \text{ attempted}$$

M1

All correct

A1

$$\left(\int (\ln x)^2 dx = \right) x(\ln x)^2 - \int x \times \frac{2}{x} \ln x (dx)$$

OE correct substitution of their terms into parts

m1

$$= x(\ln x)^2 - 2(x \ln x - x) + C \quad \text{OE}$$

All correct (constant needed) including correct use of brackets. **Do not penalise missing constant if already penalised in part (i)**

ISW

A1

4

(b) $\frac{du}{dx} = \frac{1}{2\sqrt{x}} \quad \text{or} \quad \frac{1}{2}x^{-\frac{1}{2}}$

$$(dx = 2u \, du)$$

$$u = \sqrt{x}$$

B1

$$\int_{(1)}^{(4)} \frac{1}{x + \sqrt{x}} dx = \int_{(1)}^{(2)} \frac{1}{u^2 + u} 2u \, (du)$$

All in terms of u including attempt at replacing dx (not simply writing du), condone missing limits and du

M1

Integrand correct unsimplified

A1

$$= 2 \int_{(1)}^{(2)} \frac{1}{u+1} (du)$$

A1

$$= 2 \ln(u+1) \Big|_{(1)}^{(2)}$$

FT their $\int \frac{k}{u+1} (du)$

A1F

$$= 2 \ln(2+1) - 2 \ln(1+1) \text{ or } 2 \ln(\sqrt{4}+1) - 2 \ln(\sqrt{1}+1)$$

correct use of correct limits on $k \ln(u+1)$ or $k \ln(\sqrt{x}+1)$

A1F

$$= 2 \ln \frac{3}{2} \text{ or } \ln \frac{9}{4} \text{ or } 2 \ln 3 - 2 \ln 2$$

OE ISW

A1

7

[15]

Q9.

(a) $\int t \cos\left(\frac{\pi}{4}t\right) dt$

Clear attempt to use parts

$$u = t \quad \frac{dv}{dt} = \cos\left(\frac{\pi}{4}t\right)$$

$$\frac{du}{dt} = 1 \quad v = \frac{4}{\pi} \sin\left(\frac{\pi}{4}t\right)$$

M1

$$= t \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}t\right) - \frac{4}{\pi} \int \sin\left(\frac{\pi}{4}t\right) (dt)$$

Must be in terms of π

A1

$$= \frac{4}{\pi} t \sin\left(\frac{\pi}{4}t\right) + \frac{4}{\pi} \cos\left(\frac{\pi}{4}t\right)$$

Correct form, any non-zero values for p, q

m1

$$= t \times \frac{4}{\pi} \sin\left(\frac{\pi t}{4}\right) + \frac{4}{\pi} \times \frac{4}{\pi} \cos\left(\frac{\pi t}{4}\right)$$

Any correct unsimplified form.

Constant not required

A1

4

(b) $\int 32x \, dx = \int t \cos\left(\frac{\pi t}{4}\right) dt$

Correct separation and notation.

B1

$$16x^2 =$$

$$\frac{x^2}{2} \text{ if 32 not brought over; allow } 32 \times \frac{x^2}{2}$$

B1

$$t \times \frac{4}{\pi} \sin\left(\frac{\pi t}{4}\right) + \frac{16}{\pi^2} \cos\left(\frac{\pi t}{4}\right) + C$$

Equate to result from part (a) with constant and use (0,4) to find a value for the constant

M1

$$C = 256 - \frac{16}{\pi^2}$$

Accept $C = 254$ or better (254.37886...)

A1

$$t = 45$$

$$16x^2 = -40.514... - 1.146... + 254.378... = 212.718...$$

$$x^2 = 13.294...$$

$$x = 3.646... = 3.65 \text{ m or (height =) } 365 \text{ cm}$$

$$\text{Substitute } t = 45 \text{ into } kx^2 = pt \sin\left(\frac{\pi t}{4}\right) + q \cos\left(\frac{\pi t}{4}\right) + C$$

$p \neq 0$, $q \neq 0$ and calculate x .

CSO

m1A1

6

[10]

Q10.

(a) $\left(\frac{dy}{dx}\right) 12x^2 - 6$
do not ISW

B1

1

(b) $\int_2^3 \frac{2x^2 - 1}{4x^3 - 6x + 1} dx = \left[\frac{1}{6} \ln(4x^3 - 6x + 1) \right]_{(2)}^{(3)}$
 $k \ln(4x^3 - 6x + 1)$, k is a constant

M1

$$k = \frac{1}{6}$$

A1

$$= \frac{1}{6} \ln(4 \times 3^3 - 6 \times 3 + 1) - \frac{1}{6} \ln(4 \times 2^3 - 6 \times 2 + 1)$$

correct substitution in $F(3) - F(2)$.
condone poor use or lack of brackets.

m1

$$= \frac{1}{6} \ln 91 - \frac{1}{6} \ln 21$$

$k \ln 91 - k \ln 21$
only follow through on their k

A1F

$$= \frac{1}{6} \ln \frac{91}{21} \quad \text{or} \quad \left(= \frac{1}{6} \ln \frac{13}{3} \right)$$

A1

or if using the substitution

$$u = 4x^3 - 6x + 1$$

$$\int = k \int \frac{du}{u} \quad \text{M1}$$

$$= \frac{1}{6} \ln u \quad \text{A1}$$

then, either change limits to 21 and 91 m1 then A1F A1
as scheme or changing back to 'x', then m1 A1F A1 as scheme

5

[6]

Q11.

(a) $\left(\frac{dx}{d\theta} = \right) \frac{(\sin \theta \times 0) - 1 \times \cos \theta}{\sin^2 \theta}$
 quotient rule $\frac{\pm \sin \theta \times k \pm 1 \times \cos \theta}{\sin^2 \theta}$ where $k = 0$ or 1

M1

must see the '0' either in the quotient or in eg $\frac{dx}{d\theta} = 0$ etc

A1

$$= -\frac{\cos \theta}{\sin^2 \theta} \quad \text{or} \quad = -\frac{\cos \theta}{\sin \theta \sin \theta}$$

or equivalent

$$= -\operatorname{cosec} \theta \cot \theta$$

CSO, AG must see one of the previous expressions

A1

(b) $x = \operatorname{cosec} \theta$

$$\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta$$

OE, eg $dx = -\operatorname{cosec} \theta \cot \theta d\theta$

B1

Replacing $\sqrt{(\operatorname{cosec}^2 \theta - 1)}$ by $\sqrt{\cot^2 \theta}$, or better
 at any stage of solution

B1

$$\int = \int \frac{-\operatorname{cosec} \theta \cot \theta}{\operatorname{cosec}^2 \theta \sqrt{(\operatorname{cosec}^2 \theta - 1)}} d\theta$$

all in terms of θ , and including their attempt at dx ,
 but condone omission of $d\theta$

M1

fully correct and must include $d\theta$ (at some stage in solution)

A1

$$\int \frac{-\operatorname{cosec} \theta \cot \theta}{\operatorname{cosec}^2 \theta \cot \theta} (d\theta) = \int \frac{-1}{\operatorname{cosec} \theta} (d\theta)$$

OE eg $\int -\sin \theta (d\theta)$

A1

$$= \cos \theta$$

A1

$$x = 2, \theta = 0.524 \text{ AWRT}$$

$$x = \sqrt{2}, \theta = 0.785 \text{ AWRT}$$

$$\text{correct change of limits or } (\pm) \cos \theta = (\pm) \left[\sqrt{1 - \frac{1}{x^2}} \right]_{\sqrt{2}}^2 \text{ OE}$$

B1

$$0.8660 - 0.7071$$

$$\text{c's } F(0.52) - F(0.79)$$

$$\text{substitution into } \pm \cos \theta \text{ only or } \left(\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \right)$$

m1

$$= 0.159$$

A1

9

[12]

Q12.

$$\int \frac{dy}{y^2} = \int x \sin 3x \, dx$$

*Correct separation and notation;
condone missing integral signs*

B1

$$\int \frac{dy}{y^2} = -\frac{1}{y}$$

B1

$$\int x \sin 3x \, dx = x \left(-\frac{1}{3} \cos 3x \right)$$

$$u = x \quad \frac{dv}{dx} = \sin 3x$$

$$\frac{du}{dx} = 1 \quad v = k \cos 3x$$

Use parts formula with correct substitution into

$$- \int -\frac{1}{3} \cos 3x$$

A1

$$= -\frac{1}{3}x \cos 3x \, dx + \frac{1}{9} \sin 3x$$

CAO

A1

$$-\frac{1}{y} = -\frac{1}{3}x \cos 3x + \frac{1}{9} \sin 3x + C$$

$$-1 = -\frac{1}{3} \times \frac{\pi}{6} \cos\left(\frac{\pi}{2}\right) + \frac{1}{9} \sin\left(\frac{\pi}{2}\right) + C$$

$$\text{Use } x = \frac{\pi}{6} \, y = 1 \text{ to find } C$$

M1

$$C = -\frac{10}{9}$$

CAO

A1

$$-\frac{1}{y} = -\frac{1}{9} (3x \cos 3x \, dx - \sin 3x + 10)$$

$$\text{And invert to } -y = -\frac{9}{(\dots)}$$

m1

$$y = \frac{9}{3x \cos 3x - \sin 3x + 10}$$

CSO, condone first B1 not given

A1

Second M1 finding C; substitute $x = \frac{\pi}{6} \, y = 1$ into $f(y) = px \cos 3x + q \sin 3x + C$ and evaluate using radians.

Must calculate a value of C.

$$\text{m1 for reaching form } \pm \frac{k}{y} = \frac{1}{9} (Px \cos 3x + Q \sin 3x + R) \text{ where } P \text{ and } Q \text{ are } \pm 3 \text{ or } \pm \frac{1}{3}$$

$$\text{or } \pm 1 \text{ and inverting to } \pm \frac{y}{k} = \frac{9}{(Px \cos 3x + Q \sin 3x + R)}$$

[9]

Q13.

$$u = x^4 + 2$$

$$\frac{du}{dx} = 4x^3$$

$$\text{or } du = 4x^3 dx$$

B1

$$\int \frac{x^7}{(x^4 + 2)^2} dx = \int \frac{k(u-2)}{u^2} du \quad \text{or} \quad \int \frac{k(u-2)^{\frac{7}{4}}}{u^2} \frac{du}{(u-2)^{\frac{3}{4}}}$$

Either expression all in terms of u including replacing dx , but condone omission of du

M1

$$= \left(\frac{1}{4}\right) \int \frac{1}{u} - \frac{2}{u^2} du$$

$$k \int au^{-1} + bu^{-2} du, \text{ where } k, a, b \text{ are constants}$$

m1

$$= \left(\frac{1}{4}\right) \left[\ln u + \frac{2}{u} \right]$$

Must have seen du on an earlier line where every term is a term in u

A1

$$= \left(\int = \left(\frac{1}{4}\right) \left[\ln u + \frac{2}{u} \right]_2^3 \right)$$

$$\left(\left(\frac{1}{4}\right) \left[\ln(x^4 + 2) + \frac{2}{(x^4 + 2)} \right]_0^1 \right)$$

$$= \left(\frac{1}{4}\right) \left[\left(\ln 3 + \frac{2}{3} \right) - (\ln 2 + 1) \right]$$

Dependent on previous A1

Correct change of limits, correct substitution and $F(3) - F(2)$

or

correct replacement of u , correct substitution and $F(1) - F(0)$

m1

$$= \frac{1}{4} \ln \left(\frac{3}{2} \right) - \frac{1}{12}$$

OE in exact form

A1

[6]

Q14.

(a) $\frac{dh}{dt}$

B1

derivative = * $\times (2 - h)$

Use of $2 - h$ or $h - 2$;

* is a constant or expression in h and / or t .

M1

$$\frac{dh}{dt} = k(2 - h)$$

All correct; must be $(2 - h)$

A1

3

(b) (i) $\int x\sqrt{2x-1} \, dx = \int \frac{1}{15} \, dt$

Correct separation and notation;
condone missing integral signs.

B1

$$= \frac{1}{15} t$$

B1

Substitute $u = 2x - 1$

$$\int x\sqrt{2x-1} \, dx = \int \frac{1}{2}(u+1)\sqrt{u} \, \frac{1}{2} du$$

Suitable substitution and attempt to write integral in terms
of u including dx replaced by $\frac{1}{2} du$ or $2du$.

M1

$$= \left(\frac{1}{4}\right) \int u^{\frac{3}{2}} + u^{\frac{1}{2}} \, du$$

$\frac{1}{4}$ need not be seen

A1

$$= \frac{1}{4} \left(\frac{2}{5} u^{\frac{5}{2}} + \frac{2}{3} u^{\frac{3}{2}} \right) (+ C)$$

Integration correct including $\frac{1}{4}$

A1

$$x = 1; t = 0$$

$$u = 1, t = 0 \quad \frac{1}{4} \left(\frac{2}{5} + \frac{2}{3} \right) + C = 0$$

Use $x = 1, t = 0$ to find a value for constant C from equation in x and t .

M1

$$C = -\frac{4}{15}$$

$$C = -0.2666\dots$$

$$C = -0.267 \text{ or better}$$

A1

$$t = \frac{1}{2} \left(3(2x-1)^{\frac{5}{2}} + 5(2x-1)^{\frac{3}{2}} \right) - 4$$

$$\text{ISW } t = (2x-1)^{\frac{3}{2}} (3x+1) - 4$$

A1

Alternative (Parts)

As before

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(B1B1)

$$u = x, \quad \frac{dv}{dx} = (2x-1)^{\frac{1}{2}}$$

Attempt to use parts

(M1)

$$du = 1 \quad v = k(2x-1)^{\frac{3}{2}}$$

$$\int x\sqrt{2x-1} \, dx = x \frac{1}{3} (2x-1)^{\frac{3}{2}} - \int \frac{1}{3} (2x-1)^{\frac{3}{2}} \, dx$$

Condone missing dx

(A1)

$$= x \frac{1}{3} (2x-1)^{\frac{3}{2}} - \frac{1}{15} (2x-1)^{\frac{5}{2}} (+C)$$

(A1)

$$x = 1, t = 0 \quad \frac{1}{3} - \frac{1}{15} + C = 0$$

Use $x = 1, t = 0$ to find a value for constant C from equation in x and t

(M1)

$$C = -\frac{4}{15}$$

$$C = -0.2666\dots$$

$$C = -0.267 \text{ or better}$$

(A1)

$$t = 5x (2x-1)^{\frac{3}{2}} - (2x-1)^{\frac{5}{2}} - 4$$

$$\text{ISW } t = (2x-1)^{\frac{3}{2}} (3x+1) - 4$$

(A1)

8

(ii) $x = 2 \quad t = 32.4 \quad (\text{minutes})$
 32.4 or better (32.373...)

B1

1

[12]

Q15.

(a) (i) $\int \frac{1}{3+2x} dx$

$$= k \ln(3+2x)$$

Where k is a rational number

M1

$$= \frac{1}{2} \ln(3+2x) + c$$

Or

if substitution $u = 3+2x, du = 2dx$

$$\int = \int \frac{1}{u} \frac{du}{2} = k \ln u$$

M1

$$= \frac{1}{2} \ln(3 + 2x) + c$$

A1

A1

2

(b) $u = x \quad dv = \sin \frac{x}{2}$

$$\int \sin \frac{x}{2} (dx) = k \cos \frac{x}{2}, \frac{d}{dx}(x) = 1$$

where k is a constant

M1

$$du = 1 \quad v = -2 \cos \frac{x}{2}$$

All correct

A1

$$\int = -2x \cos \frac{x}{2} - \int -2 \cos \frac{x}{2} (dx)$$

Correct substitution of their terms into parts formula (watch signs carefully)

m1

$$= -2x \cos \frac{x}{2} + 4 \sin \frac{x}{2} + c$$

CAO

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A1

4

[6]

Q16.

$$\int \frac{1}{\cos^2 x (1 + 2 \tan x)^2} dx$$

$$u = 1 + 2 \tan x$$

$$\left(\frac{du}{dx} = \right) 2 \sec^2 x \text{ OE}$$

condone $\left(\frac{du}{dx} = \right) a \sec^2 x$ where a is a constant

M1

$$\int = \int \frac{du}{2u^2}$$

$$\int \frac{k}{u^2} (du), \text{ where } k \text{ is a constant}$$

m1

correct, or $\frac{1}{2} \int u^{-2} (du)$

A1

$$= \frac{1}{2} u^{-1}$$

correct integral of their expression but must have scored M1 m1

A1F

$$= -\frac{1}{2u}$$

$$= -\frac{1}{2(1+2\tan x)} (+c)$$

CSO, no ISW

A1

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[5]

Q17.

(a) $\int x \sin(2x - 1) dx$

$$u = x \quad \frac{dv}{dx} = \sin(2x - 1)$$

$$\int \sin f(x), \frac{d}{dx}(x) \text{ attempted}$$

M1

$$\frac{du}{dx} = 1 \quad v = -\frac{1}{2} \cos(2x - 1)$$

All correct – condone omission of brackets

A1

$$\left(\int = \right) - \frac{x}{2} \cos(2x-1) - \int -\frac{1}{2} \cos(2x-1) (dx)$$

correct substitution of their terms into parts

m1

$$= -\frac{x}{2} \cos(2x-1) + \frac{1}{2} \int \cos(2x-1) (dx)$$

All correct – condone omission of brackets

A1

$$= -\frac{x}{2} \cos(2x-1) + \frac{1}{4} \sin(2x-1) + c$$

CSO condone missing + c and dx

Condone missing brackets around 2x – 1 if recovered in final line ISW

A1

5

(b) $u = 2x - 1$
 $'du = 2 dx'$

OE

M1

$$\int \frac{x^2}{2x-1} dx = \int \frac{(u+1)^2}{4u} \frac{du}{2}$$

All in terms of u

m1

All correct

A1

PI from later working

$$= \left(\frac{1}{8} \right) \int \frac{u^2 + 2u + 1}{u} du$$

$$= \left(\frac{1}{8} \right) \int u + 2 + \frac{1}{u} du$$

A1

$$= \left(\frac{1}{8}\right) \left[\frac{u^2}{2} + 2u + \ln u \right]$$

$$\text{or } \left(\frac{1}{8}\right) \left[\frac{(u+2)^2}{2} + \ln u \right]$$

B1

$$= \frac{1}{8} \left[\frac{(2x-1)^2}{2} + 2(2x-1) + \ln(2x-1) \right] + c$$

$$\text{or } = \frac{1}{8} \left[\frac{(2x+1)^2}{2} + \ln(2x-1) \right] + c$$

CSO condone missing + c only ISW

A1

6

[11]

Q18.

(a) $y = \frac{4x}{4x-3}$

$$\frac{dy}{dx} = \frac{(4x-3) \cdot 4 - 4x(4)}{(4x-3)^2}$$

Must use quotient rule
Condone one slip

M1

$$= \frac{-12}{(4x-3)^2}$$

$$k = -12$$

A1

2

(b) (i) $y = x \ln(4x-3)$

$$\frac{dy}{dx} = \frac{x \cdot 4}{4x-3} + \ln(4x-3)$$

$$\frac{f'(x)}{4x-3} + g(x) \quad \text{'f'(x)' may be constant}$$

M1

$$\frac{kx}{4x-3} + \ln(4x-3)$$

m1A1

3

(ii) $x = 1 \quad y = 0$

B1

$$\frac{dy}{dx} = 4$$

Sub $x = 1$ into their $\frac{dy}{dx}$

M1

$\therefore y = 4(x - 1)$ any correct form
CSO Must have full marks in (b)(i)

A1

3

(c) (i) $u = 4x - 3$
 $du = 4 dx$

M1

$$\int \frac{4x}{4x-3} dx = \int \frac{u+3}{u} \frac{du}{4}$$

Or $\int \frac{4x}{4x-3} dx = \int \left(1 + \frac{3}{4x-3}\right) dx$

A1

$$= \left(\frac{1}{4}\right) \int \left(1 + \frac{3}{u}\right) (du)$$

$$= \int \left(1 + \frac{3}{u}\right) du \text{ etc}$$

m1

$$= \frac{1}{4} (u + 3 \ln u)$$

$$= \frac{1}{4} [(4x - 3) + 3 \ln (4x - 3)] (+c)$$

CSO Condone missing du

A1

4

(ii) $\int \ln(4x - 3) \, dx$

$$u = \ln(4x - 3) \quad \frac{dv}{dx} = 1$$

In correct direction

M1

$$\frac{du}{dx} = \frac{4}{4x - 3} \quad v = x$$

$$\int x \ln(4x - 3) - \int \frac{4x}{4x - 3} \, dx$$

A1

$$= x \ln(4x - 3) - \frac{1}{4} [(4x - 3) + 3 \ln(4x - 3)]$$

x ln(4x - 3) - their (c)(i)

m1A1

(+ c)

4

[16]

Q19.

(a) $\int (t - 1) \ln t \, dt$

$$u = \ln t \quad \frac{dv}{dt} = t - 1$$

Differentiate + integrate, correct direction

M1

$$\frac{du}{dt} = \frac{1}{t} \quad v = \frac{t^2}{2} - t$$

All correct

A1

$$\begin{aligned} \int &= \left(\frac{t^2}{2} - t \right) \ln t - \left(\frac{t^2}{2} - t \right) \times \frac{1}{t} (dt) \\ &= \left(\frac{t^2}{2} - t \right) \ln t - \int \left(\frac{t}{2} - 1 \right) (dt) \end{aligned}$$

Condone missing brackets

A1

$$= \left(\frac{t^2}{2} - t \right) \ln t - \frac{t^2}{4} + t + c$$

CAO

A1

4

Alternative

$$\int (t-1) \ln t$$

$$u = \ln t \quad v' = (t-1)$$

(M1)

$$u' = \frac{1}{t} \quad v = \frac{(t-1)^2}{2}$$

(A1)

$$\begin{aligned} \int &= \frac{(t-1)^2}{2} \ln t - \int \frac{(t-1)^2}{t} \frac{1}{t} dt \\ &= \frac{(t-1)^2}{2} \ln t - \frac{1}{2} \int \frac{t^2 - 2t + 1}{t} dt \\ &= \frac{(t-1)^2}{2} \ln t - \frac{1}{2} \int \left(t - 2 + \frac{1}{t} \right) dt \end{aligned}$$

(A1)

$$\frac{(t-1)^2}{2} \ln t - \frac{1}{2} \left[\frac{t^2}{2} - 2t + \ln t \right]$$

(A1)

$$\begin{aligned} &= \frac{t^2}{2} \ln t - t \ln t + \frac{1}{2} \ln t - \frac{t^2}{4} + t - \frac{1}{2} \ln t \\ &= \left(\frac{t^2}{2} - t \right) \ln t - \frac{1}{4} t^2 + t + c \end{aligned}$$

(4)

(b) $t = 2x + 1$
 $dt = 2 dx$ (RHS)

$$\frac{dt}{dx} = 2 \quad (LHS)$$

M1

$$2x = t - 1,$$

OE

m1

$$\int = \int 2(t-1) \ln t \frac{dt}{2}$$

AG

A1

3

(c) $\begin{bmatrix} x \end{bmatrix}_b = \begin{bmatrix} t \end{bmatrix}_1$

Limit becoming 3

M1

$$\begin{aligned} \int &= \left[\left(\frac{t^2}{2} - t \right) \ln t - \frac{t^2}{4} + t \right]_1^3 \\ &= \left[\left(\frac{9}{2} - 3 \right) \ln 3 - \frac{9}{4} + 3 \right] - \left[0 - \frac{1}{4} + 1 \right] \end{aligned}$$

Correctly sub. 1, 3 into their (a)

m1

$$= \frac{3}{2} \ln 3$$

CSO

A1

3

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or

$$\int = \left[\left(\frac{(2x+1)^2}{2} - (2x+1) \right) \ln (2x+1) - \frac{(2x+1)^2}{4} + (2x+1) \right]_0^1$$

Condone 1 slip

(M1)

$$= \left(\left(\frac{9}{2} - 3 \right) \ln 3 - \frac{9}{4} + 3 \right) - \left(0 - \frac{1}{4} + 1 \right)$$

Correctly sub. 0, 1

(m1)

$$= \frac{3}{2} \ln 3$$

CSO

(A1)

(3)

[10]

Q20.

(a) (i) $y = 2x^2 - 8x + 3$

$$\left(\frac{dy}{dx}\right) = 4x - 8$$

B1

1

(ii)
$$\int_4^6 \frac{x-2}{2x^2-8x+3} dx$$

$$= \frac{1}{4} [\ln |2x^2 - 8x + 3|]_4^6$$

M1 for $k \ln (2x^2 - 8x + 3)$; allow $k \ln u$

M1A1

$$= \frac{1}{4} [\ln 27 - \ln 3]$$

*Correct substitution into
 $k \ln (2x^2 - 8x + 3)$ or 3, 27 into $k \ln u$*

m1

$$= \frac{1}{4} \ln 9$$

$$= \frac{1}{2} \ln 3$$

A1

4

(b)
$$\int x\sqrt{3x-1} dx$$

$$u = 3x - 1 \quad du = 3 dx$$

OE

B1

$$\int = \left(\frac{1}{9}\right) \int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}}\right) (du)$$

[2 terms in with rational indices

M1

$$= \left(\frac{1}{9}\right) \left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{u^{\frac{3}{2}}}{\frac{3}{2}} (+c) \right]$$

Must be 2 terms with correct indices

$$\left(\text{only ft for } x = \frac{u-1}{3} \right)$$

A1F

$$= \frac{2}{45} (3x-1)^{\frac{5}{2}} + \frac{2}{27} (3x-1)^{\frac{3}{2}} + c$$

CSO OE

A1

4

[9]

Q21.

(a) $\frac{dy}{dx} = -x2 \sin 2x + \cos 2x$

product rule $kx \sin 2x \pm \cos 2x$

M1

no further incorrect working

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A1

2

(b) (i) $-2\alpha \sin 2\alpha + \cos 2\alpha = 0$

replacing $x = \alpha$ and writing equation equal to zero (at any line)

M1

$$\left. \begin{array}{l} 2\alpha \sin 2\alpha = \cos 2\alpha \\ 2\alpha \tan 2\alpha = 1 \end{array} \right\} \text{either}$$

$$2\alpha \tan 2\alpha - 1 = 0$$

AG; CSO

A1

2

(ii) $\left. \begin{array}{l} f(0.4) = 0.2 \\ f(0.5) = -0.6 \end{array} \right\} \text{awrt} \quad \text{o.e}$

(0.9's unsubstantiated scores M0)

M1

Change of sign $\therefore 0.4 < \alpha < 0.5$

A1

2

(iii) $2x \tan 2x = 1$

$$\left. \begin{array}{l} \tan 2x = \frac{1}{2x} \\ 2x = \tan^{-1}\left(\frac{1}{2x}\right) \end{array} \right\} \text{either}$$

$$x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$$

AG; CSO

B1

1

(iv) $x_1 = 0.4$
 $x_2 = 0.4480...$
 $x_2 = 25.7$

$x_3 = 0.4200...$
 $= 0.42$

M1

A1

2

(c) $y = x \cos 2x$

$$\left. \begin{array}{l} u = x \quad \frac{du}{dx} = 1 \\ \frac{dv}{dx} = \cos 2x \quad v = \frac{\sin 2x}{2} \end{array} \right\}$$

$\left. \begin{array}{l} \text{differentiate one term} \\ \text{integrate one term} \end{array} \right\} \text{ must be } k \sin 2x$

M1

*correct substitution of their values into
 parts formula using $u = x$*

m1

$$\begin{aligned}
 \int &= \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{4} (dx) \\
 &= \left[\frac{x \sin 2x}{2} + \frac{\cos 2x}{4} \right]_{(0)}^{(0.5)}
 \end{aligned}$$

A1

$$= \left(\frac{\sin 1}{4} + \frac{\cos 1}{4} \right) - \left(\frac{\cos 0}{4} \right)$$

*correctly substituting values from
previous 2 method marks*

m1

$$= 0.0954$$

AWRT

A1

5

[14]

Q22.

(a) $y = \frac{\sin \theta}{\cos \theta}$

$$\frac{dy}{d\theta} = \frac{\cos \theta \cos \theta - \sin \theta (-\sin \theta)}{\cos^2 \theta}$$

$$\frac{\pm \cos^2 \theta \pm \sin^2 \theta}{\cos^2 \theta}$$

$$= \frac{1}{\cos^2 \theta} \quad \text{o.e.}$$

$$(1 + \tan^2 \theta)$$

$$= \sec^2 \theta$$

AG; CSO

A1

Alternative:

$$y = \frac{\tan \theta}{1}$$

$$\frac{dy}{d\theta} = \frac{1 \sec^2 \theta - 0}{1^2}$$

M1A1

$$= \sec^2 \theta$$

A1

3

(b)

$$x = \sin \theta$$

$$x^2 = \sin^2 \theta$$

$$\cos^2 \theta = 1 - x^2$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{x}{\sqrt{1-x^2}}$$

OR LHS =

$$\frac{\sin \theta}{\sqrt{1-\sin^2 \theta}}$$

$$= \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta \quad \text{AG}$$

use of $\cos^2 \theta + x^2 = 1$ M1

AG; CSO

A1

2

(c)

$$\int \frac{1}{(1-x^2)^{\frac{3}{2}}} dx$$

$$x = \sin \theta$$

$$dx = \cos \theta d\theta$$

$$\frac{dx}{d\theta} = \pm \cos \theta$$

o.e.

M1

$$\int = \int \frac{\cos \theta (d\theta)}{(1-\sin^2 \theta)^{\frac{3}{2}}}$$

all in terms of θ

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m1

$$= \int \frac{\cos \theta}{(\cos^2 \theta)^{\frac{3}{2}}} (d\theta)$$

A1

$$= \int \sec^2 \theta (d\theta)$$

A1

$$= \tan \theta$$

$$= \frac{x}{\sqrt{1-x^2}} (+c)$$

CSO including dθ's

A1

5

[10]

Q23.

(a) $\int x e^{5x} dx$

$u = x \quad dv = e^{5x}$

integrate one term, differentiate one term

M1

$du = 1 \quad v = \frac{1}{5} e^{5x}$

A1

$\int = \frac{1}{5} x e^{5x} - \int \frac{1}{5} e^{5x} dx$

A1

$= \frac{1}{5} x e^{5x} - \frac{1}{25} e^{5x} (+c)$

A1

4

(b) (i) $u = x^{\frac{1}{2}}$

$du = \frac{1}{2} x^{-\frac{1}{2}} dx$

M1

$\int = \int \frac{1}{1+u} \times 2 du$

correct with no errors; AG

A1

2

(ii) $\int_1^9 dx = \int_1^3 \frac{2}{1+u} du$

correct limits used in correct expression, ignoring k

m1

$$= [2 \ln(1+u)]_1^3$$

for k ln (1 + u)

M1

$$= 2 \ln 4 - 2 \ln 2$$

$$(\ln 4)$$

ISW OE

A1

3

[9]

Q24.

(a) (i) $f' = \frac{dy}{dx} = 4x^3 + 2$

B1

1

(ii) $\int \frac{2x^3 + 1}{x^4 + 2x} dx$

$$= \frac{1}{2} \ln(x^4 + 2x) + c$$

For k ln (x⁴ + 2x)

M1

By substitution k ln u M1
correct A1

A1

2

(b) (i) $u = 2x + 1$

$$du = 2 dx$$

B1

$$\int x\sqrt{2x+1} dx =$$

$$\int \left(\frac{u-1}{2}\right)\sqrt{u} \frac{du}{2}$$

Must be in terms of u only incl. du

M1

$$= \frac{1}{4} \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du$$

AG

A1

3

(ii) $\int_0^4 dx = \int_1^9 du$

Or changing u's to x's at end

B1

$$\frac{1}{4} \int u^{\frac{3}{2}} - u^{\frac{1}{2}} = \frac{1}{4} \left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} - \frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right]$$

M1

A1

$$= \frac{1}{4} \left[\left(\frac{2}{5} (9)^{\frac{5}{2}} - \frac{2}{3} (9)^{\frac{3}{2}} \right) - \left(\frac{2}{5} - \frac{2}{3} \right) \right]$$

$$= \frac{1}{4} [79.2 + 0.26]$$

$$= 19.86$$

Sight of any of these 3 lines

$$= 19.9$$

AG

A1

4

[10]

Q25.

(a) $y = x^{-2} \ln x$

$$\frac{dy}{dx} = x^{-2} \frac{1}{x} - 2x^{-3} \ln x$$

Use of product or quotient

M1

each term

A1 A1

$$= \frac{1 - 2 \ln x}{x^3}$$

Convincing argument $x^{-2} \times \frac{1}{x} = x^{-3}$
 AG

A1

4

(b) $\int x^{-2} \ln x \, dx$ $u = \ln x$ $dv = x^{-2}$
Attempt at integration by parts

M1

$$du = \frac{1}{x} \quad v = -x^{-1}$$

A1

$$\int = -\frac{1}{x} \ln x + \int x^{-2} \, dx$$

A1

$$= -\frac{1}{x} \ln x - \frac{1}{x} (+c)$$

A1

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4

(c) (i) At A, $\frac{dy}{dx} = 0$

$$1 - 2 \ln x = 0$$

$$\ln x = \frac{1}{2}$$

Attempt at $\ln x = k$

M1

$$x = e^{\frac{1}{2}}$$

A1

$$(ii) \quad R = \left[-\frac{1}{x} (\ln x + 1) \right]_1^5$$

$$R = [\text{Their (b)}]_1^5$$

M1

$$= -\frac{1}{5} (\ln 5 + 1) + (\ln 1 + 1)$$

OE

A1

$$= \frac{1}{5} (4 - \ln 5)$$

convincing argument; AG

A1

3

[13]

Q26.

$$(a) \quad y = (3x-1)^{10}$$

$$\frac{dy}{dx} = 10 (3x-1)^9 \times 3$$

M1 for $a(3x-1)^9$ where $a = \text{constant}$

M1A1

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2

$$= 30 (3x-1)^9$$

$$(b) \quad \int x(2x+1)^8 dx$$

$$u = 2x+1$$

$$du = 2 dx$$

OE

B1

$$\int = \int \left(\frac{u-1}{2} \right) u^8 \left(\frac{du}{2} \right)$$

all in terms of u. Condone omission of du

M1

$$= \frac{1}{4} \int u^9 - u^8 \, du$$

$$= \frac{1}{4} \left[\frac{u^{10}}{10} - \frac{u^9}{9} \right]$$

$$p \frac{u^{10}}{10} + q \frac{u^9}{9}$$

B1

$$= \frac{(2x+1)^{10}}{40} - \frac{(2x+1)^9}{36} (+c)$$

OE; CAO

SC: correct answer, no working/parts in x (B1)

A1

4

[6]

Q27.

(a) $\frac{1}{4} e^{4x}$

B1

1

(b) $\int e^{4x} (2x+1) \, dx$
 $u = 2x+1$

$dv = e^{4x}$

by parts

$$du = 2 \qquad v = \frac{1}{4} e^{4x}$$

$$= \frac{1}{4} (2x+1)e^{4x} - \frac{1}{2} \int e^{4x} \, dx$$

Their $(uv - \int v \, du)$

M1

$$= \frac{1}{4} (2x+1) e^{4x} - \frac{1}{8} e^{4x} (+c)$$

A1

(c) $u = 1 + \ln x$
 $\frac{du}{dx} = \frac{1}{x}$ or $\frac{dx}{du} = e^{u-1}$

B1

$$\int = \int u \, du = \frac{u^2}{2} (+c)$$

in terms of u only

M1A1

$$= \frac{(1 + \ln x)^2}{2} (+c)$$

A1

[8]



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